■ Retail Data Analytics Project – Technical Documentation

Qlik Sense Cloud | ETL | Data Modeling | Dashboard Design

1. Project Overview

This document summarizes the analytical workflow developed for a retail dataset using Qlik Sense Cloud. The project follows a structured, layered approach that separates extraction, transformation, and visualization into independent stages. This methodology improves maintainability, transparency, and scalability, while reflecting real■world BI development practices.

Three analytical applications were created—one for each stage—to ensure a clean separation of responsibilities.

2. Architecture & Workflow

A. Extraction Layer

The extraction phase focuses on bringing raw data into the environment and preparing it for processing.

Key activities:

- Establishing a connection to the MySQL database.
- Performing initial checks on field formats, data types, and key consistency.
- Exporting intermediate datasets into QVD files, enabling faster reloads and modular development.

This layer ensures that all incoming data is validated and ready for transformation.

B. Transformation Layer

In this stage, the raw data is cleaned, standardized, and modeled into a structure suitable for analytics.

Main tasks:

- Cleaning and harmonizing fields across tables.
- Eliminating synthetic keys and circular references.
- Creating calculated fields and generating a Master Calendar.
- Building a star■schema■inspired model by denormalizing tables.
- Applying advanced Qlik scripting techniques such as JOIN, CONCATENATE, MAPPING LOAD, PRECEDING LOAD, CROSSTABLE.
- Storing the transformation logic in modular .qvs scripts.

This layer produces a clean, optimized model ready for visualization.

C. Visualization Layer

The final application focuses on delivering insights through an interactive dashboard.

Key components:

- Creation of master items (dimensions, measures, variables).
- Development of KPIs and dynamic charts.
- Intuitive navigation using filters, panels, and thematic sheets.
- Dashboard layout aligned with UX best practices for BI tools.

The result is a user■friendly analytical interface designed to support decision■making.

3. Tools & Technologies

• Qlik Sense Cloud – main analytics platform
• Qlik Scripting (.qvs) – ETL and data modeling logic
• QVD files – optimized storage for incremental reloads
• Visual Studio Code – script editing and version control
• Dropbox – external storage for large files and versioning

4. Key Achievements

- Clear separation of the pipeline into three independent layers.
- Well■documented ETL process with modular scripts.
- Data quality validation through exploratory visualizations.
- Dashboard designed with consistency, readability, and scalability in mind.
- Use of variables and master items to support dynamic, maintainable visualizations.

5. Summary

This project demonstrates a complete analytical workflow using Qlik Sense Cloud, emphasizing clean architecture, efficient data modeling, and user■centered dashboard design. The layered approach reflects industry standards and provides a solid foundation for scalable BI solutions.